NYC Transit Optimization & Anomaly Challenge

LHC Documentation

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1. Introduction

The NYC Transit Optimization & Anomaly Challenge focuses on analyzing New York City taxi trip data to understand travel patterns, predict surge levels, estimate congestion-related scores, and identify geographic route patterns.

In this project, Python and machine learning techniques are used to process taxi trip records and perform three major tasks: classification, regression, and unsupervised learning.

The outputs of all three tasks are combined into one integrated dataset for further analysis.

2. Problem Statement

Taxi trip data contains useful information about pickup locations, drop-off locations, trip distance, travel time, passenger count, and fare amount.

Analyzing this data can help us understand travel patterns and identify unusual or less frequent routes.

The main problem is to use data analysis and machine learning techniques to study taxi trips and generate useful predictions from the available dataset.

3. Objectives

1. Load and preprocess NYC taxi trip data.
2. Analyze important features and travel patterns.
3. Develop a classification model for surge tier prediction.
4. Develop a regression model for CEPI score prediction.
5. Apply clustering to identify geographic route patterns.
6. Evaluate the models and clustering results.
7. Integrate the outputs of all three pillars.
8. Save the final integrated predictions.

4. Dataset Description

The project uses NYC Yellow Taxi trip data from four CSV files:

- January 2015
- January 2016
- February 2016
- March 2016

The combined dataset contains approximately **47.25 million records**.

Important columns include:

- Pickup and drop-off datetime
- Passenger count
- Trip distance
- Pickup and drop-off longitude and latitude
- Fare amount
- Tip amount
- Payment type
- Total amount

For model development, a sample of approximately 200,000 records was selected, with 50,000 records from each source file.

The sampled dataset was divided into:

- Training data: 160,000 records
- Testing data: 40,000 records

5. Data Preprocessing

Data preprocessing was performed to prepare the dataset for machine learning.

The following operations were carried out:

1. Loaded the taxi trip CSV files.
2. Checked dataset shape, columns, and data types.
3. Converted pickup and drop-off columns into datetime format.
4. Removed invalid records.
5. Handled missing and infinite values.
6. Created additional features such as trip duration, pickup hour, weekday, and weekend indicator.
7. Prepared training and testing datasets.
8. Selected relevant features for each machine learning task.

6. Methodology

The project is divided into three main pillars.

Pillar A: Classification – Surge Tier Prediction

Objective: Predict the surge tier of taxi trips.

A Random Forest Classifier was used for classification.

New features such as pickup hour, weekday, weekend indicator, trip distance, and fare per mile were created.

For this experimental analysis, a synthetic surge tier target was constructed using selected feature thresholds.

Model Used: Random Forest Classifier

Parameters:

- Number of estimators: 100
- Maximum depth: 15
- Random state: 42
- Class weighting: Balanced

Evaluation Results:

- Accuracy: 99.98%
- Training records: 160,000
- Testing records: 40,000

The model successfully predicted the three constructed surge classes: 0, 1, and 2.

Pillar B: Regression – CEPI Score Prediction

Objective: Estimate a continuous CEPI score for taxi trips.

A synthetic CEPI score was constructed using passenger count, trip duration per mile, and trip distance.

A Random Forest Regressor was used to predict this score.

Model Used: Random Forest Regressor

Parameters:

- Number of estimators: 100
- Maximum depth: 15
- Minimum samples split: 5
- Minimum samples leaf: 2
- Random state: 42

Evaluation Results:

Metric	Result
MAE	0.0666
MSE	0.0622
RMSE	0.2494
R^2 Score	0.9994

The model achieved a low RMSE and a high R^2 score when predicting the constructed CEPI target.

Pillar C: Unsupervised Learning – Geographic Route Clustering

Objective: Identify geographic patterns in taxi pickup and drop-off locations.

The pickup and drop-off coordinates were used to create four geographic features:

- Pickup X coordinate
- Pickup Y coordinate
- Drop-off X coordinate
- Drop-off Y coordinate

The coordinates were converted into approximate kilometre distances using a reference location in New York City.

MiniBatchKMeans clustering was applied to group taxi trips into five clusters.

Model Used: MiniBatchKMeans

Number of Clusters: 5

Cluster Evaluation:

Silhouette Score: **0.3298**

Training Cluster Distribution:

Cluster	Number of Trips
0	73,658
1	1,538
2	72,879
3	3,298
4	5,670

Cluster 0 and Cluster 2 contained the largest number of trips. The remaining clusters represented less frequent geographic route patterns.

7. Integration of All Three Pillars

After completing the three pillars, their outputs were combined into one integrated dataset.

The integrated dataset contains:

1. Predicted Surge Tier from Pillar A.
2. Predicted CEPI Score from Pillar B.
3. Geographic Cluster from Pillar C.

The existing scaler and clustering model were used to assign test records to the five geographic clusters.

Integration Results:

Description	Records
Total test records	40,000
Surge Tier predictions	40,000
CEPI Score predictions	40,000
Geographic cluster assignments	39,352
Missing geographic clusters	648

The 648 missing cluster values resulted from records that did not pass the selected geographic coordinate filter.

8. Final Validation

The integrated dataset was checked to ensure that the results were valid.

The following checks were performed:

- Verified the total number of records.
- Checked missing values.
- Verified Surge Tier values were within classes 0, 1, and 2.
- Checked that CEPI scores were between 0 and 100.
- Verified geographic cluster values were between 0 and 4.
- Checked for duplicate records.

Validation Result: All validation checks passed. No duplicate rows were found.

9. Final Output

The final integrated dataset was successfully saved as:

File Name: final_integrated_predictions.csv

File Location: /kaggle/working/final_integrated_predictions.csv

Final Dataset Details:

Description	Value
Total Records	40,000
Total Columns	33
File Size	10.98 MB
File Format	CSV

The saved file was loaded again to verify that its shape remained 40,000 rows and 33 columns.

Screenshots :

Dataset description : Dataset shape and columns

```
# Dataset shape
# Check number of rows and columns

print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
print("Dataset Shape:", df.shape)
```

```
Number of Rows: 10906858
Number of Columns: 19
Dataset Shape: (10906858, 19)
```

Observation:

The dataset contains NYC Yellow Taxi trip records with information such as pickup and drop-off dates, passenger count, trip distance, fare amount, payment type, and geographic coordinates. The original January 2016 dataset contains 10,906,858 records and 19 columns. These features provide useful information for analyzing taxi trips and understanding travel patterns.

Preprocessing : Preprocessing output

```
       DATA PREPROCESSING SUMMARY
------------------------------------------------
Original records: 10906858
Cleaned records : 10836990
Removed records : 69868
Remaining missing values: 0
Remaining duplicates: 0

Final dataset shape: (10836990, 27)

Preprocessing completed
```

Observation:

Data preprocessing was performed to make the dataset suitable for analysis and machine learning. The date and time columns were converted into datetime format, invalid records were handled, and missing or infinite values were checked. Additional features such as trip duration, pickup hour, weekday, and weekend indicators were created. After preprocessing, the data was divided into training and testing sets containing 160,000 and 40,000 records, respectively.

Pillar A - Classification :

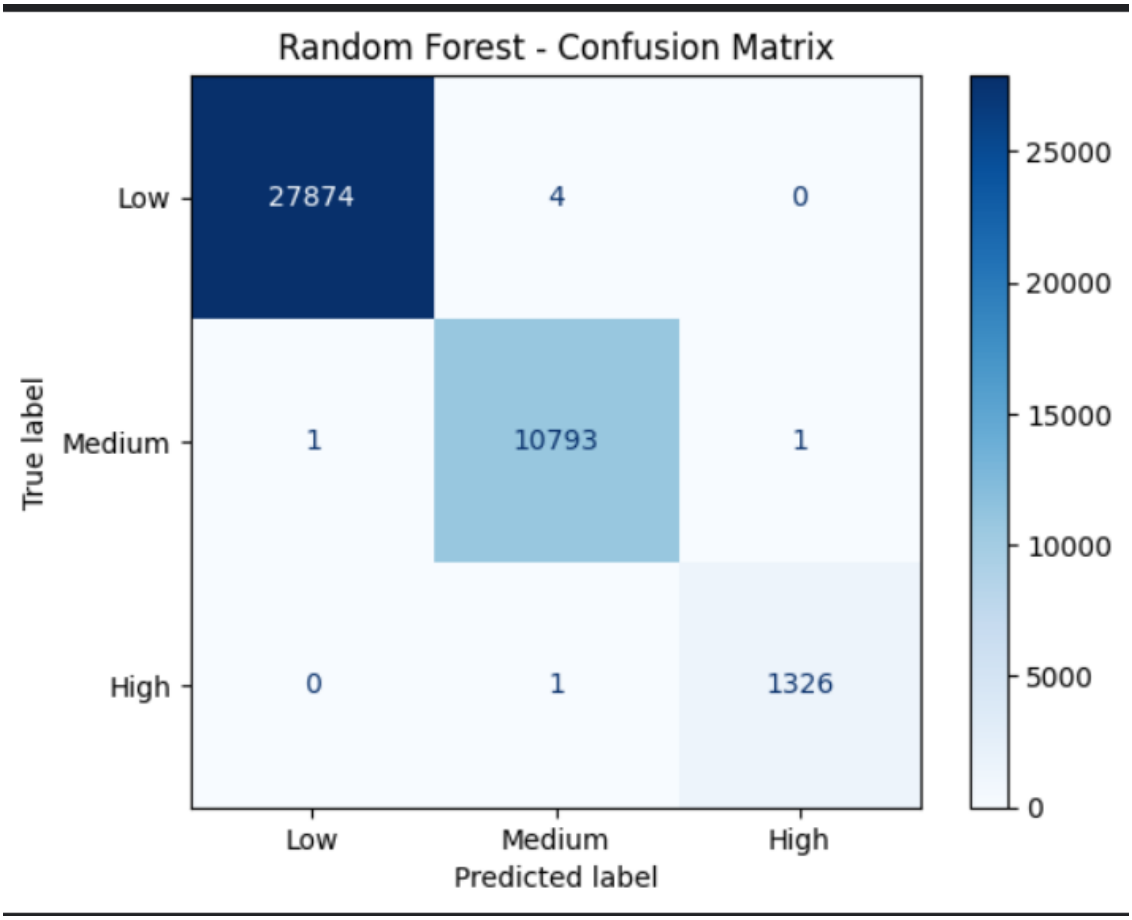
Confusion matrix and classification report :

```
Model Accuracy: 99.98 %

Classification Report:
              precision    recall  f1-score   support

   Low Surge       1.00      1.00      1.00     27878
Medium Surge       1.00      1.00      1.00     10795
  High Surge       1.00      1.00      1.00      1327

    accuracy                           1.00     40000
   macro avg       1.00      1.00      1.00     40000
weighted avg       1.00      1.00      1.00     40000
```



Observation:

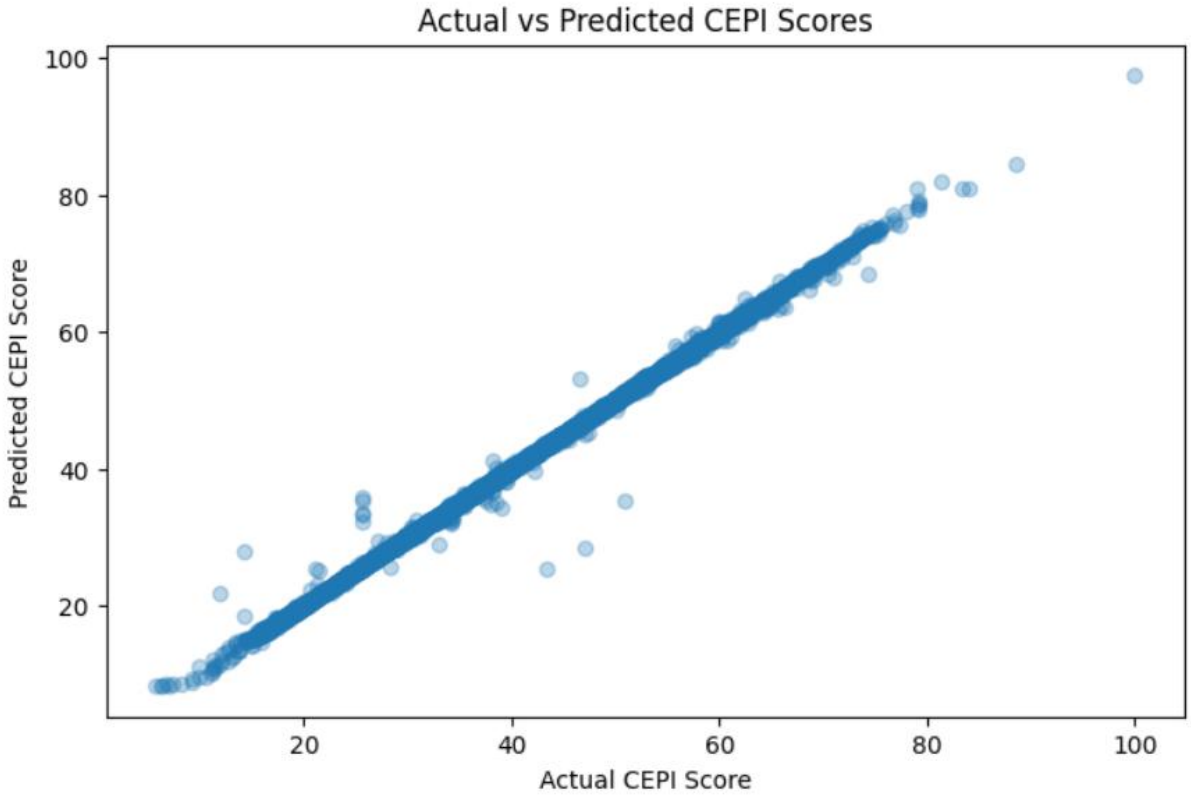
The Random Forest Classifier was used to predict three Surge Tier classes: 0, 1, and 2. The model achieved 99.98% accuracy on the test dataset. The confusion matrix shows that most predictions matched the actual constructed class labels, with only a few incorrect predictions. The classification report also shows high precision, recall, and F1-score for all three classes.

Pillar B : Actual vs predicted graph and evaluation metrics

```
CEPI predictions generated successfully
Prediction shape: (40000, 3)

First 10 predictions:
   Actual_CEPI_Score  Predicted_CEPI_Score  Prediction_Error
0          24.502995             24.495854          0.007142
1          61.068790             61.182114         -0.113325
2          30.993454             30.856823          0.136632
3          35.323130             35.275792          0.047338
4          28.598622             28.578879          0.019743
5          24.090508             23.971902          0.118605
6          30.463034             30.479232         -0.016198
7          39.182445             39.252240         -0.069795
8          28.443831             28.544954         -0.101123
9          26.193844             26.247213         -0.053369

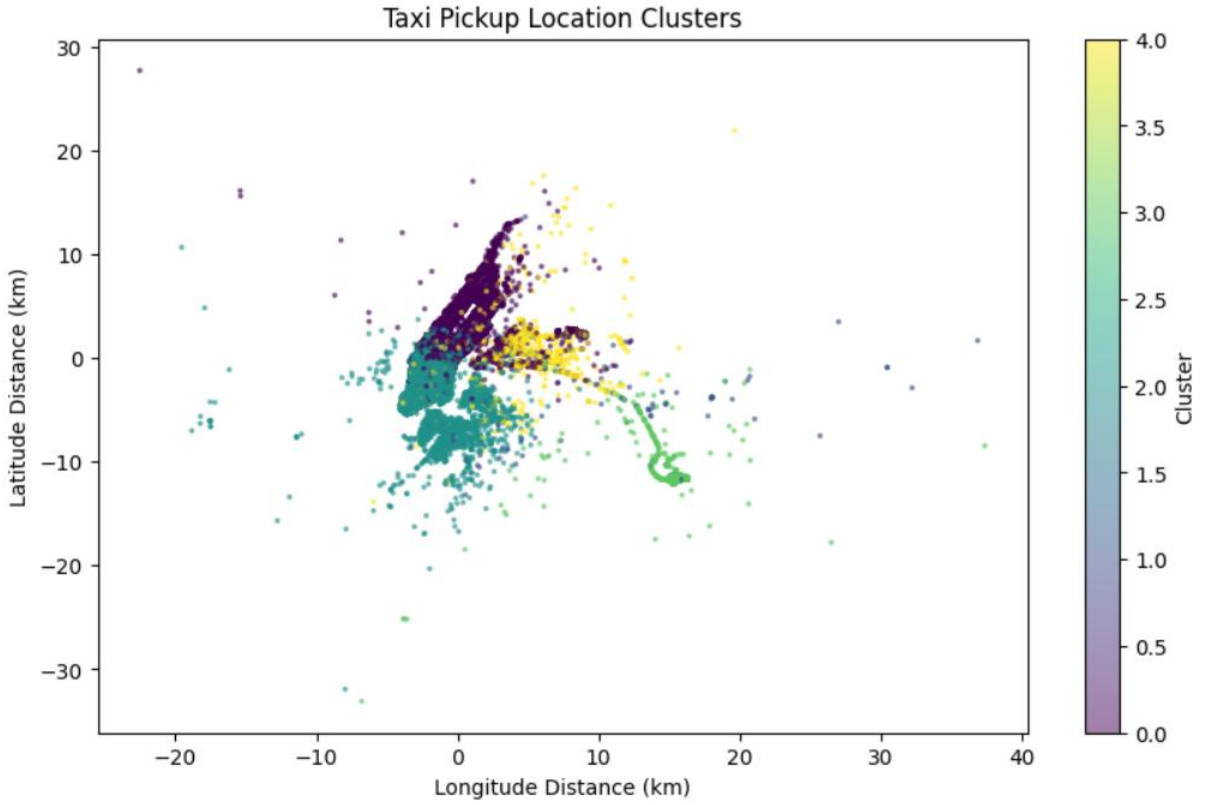
Prediction summary:
count    40000.000000
mean        38.308167
std         10.321575
min          8.247969
25%         30.505083
50%         37.106934
75%         45.208310
max         97.492026
Name: Predicted_CEPI_Score, dtype: float64
```



Observation:

The Random Forest Regressor was used to predict the CEPI score. The actual vs predicted graph shows that most predicted values are close to the actual values. The model achieved an MAE of 0.0666, RMSE of 0.2494, and R^2 score of 0.9994. These results indicate that the model closely learned the constructed CEPI score formula.

Pillar C - Cluster visualization and cluster distribution :




```
Test geographic clustering completed
Valid test records: 39352

Test cluster distribution:
cluster
0    18194
1      436
2    18369
3      856
4     1497
Name: count, dtype: int64
```

Observation:

MiniBatchKMeans clustering was applied to group taxi trips into five geographic clusters using pickup and drop-off coordinates. The visualization shows different geographic patterns represented by different colors. Cluster 0 contains 73,658 trips, while Cluster 2 contains 72,879 trips. The remaining clusters contain fewer trips and represent less frequent geographic route patterns.

Cluster evaluation :

```
# Cluster Evaluation
from sklearn.metrics import silhouette_score

# Calculate silhouette score using a sample of 5,000 records
score = silhouette_score(
    X_scaled,
    route_df["cluster"],
    sample_size=5000,
    random_state=42
)

print("Silhouette Score:", round(score, 4))
```

```
Silhouette Score: 0.3298
```

Observation:

The Silhouette Score was calculated to evaluate the quality of the five clusters. The obtained score was 0.3298, indicating moderate separation between the clusters. This means that the clustering algorithm identified some meaningful geographic patterns, although there is some overlap between the clusters.

Integration - Integrated dataset preview :

```
Pillar A columns: ['Actual_Surge_Tier', 'Predicted_Surge_Tier', 'Low_Probability', 'Medium_Probability',
'High_Probability']
Pillar B columns: ['Actual_CEPI_Score', 'Predicted_CEPI_Score', 'Prediction_Error']

Integration of all three pillars completed!
Integrated dataset shape: (40000, 33)

Integrated dataset preview:
```

	predicted_surge_tier	predicted_cepi_score	geographic_cluster
0	0	24.495854	NaN
1	0	61.182114	0.0
2	1	30.856823	2.0
3	0	35.275792	0.0
4	0	28.578879	0.0
5	0	23.971902	2.0
6	0	30.479232	0.0
7	0	39.252240	0.0
8	0	28.544954	2.0
9	1	26.247213	0.0

```
Missing geographic cluster values: 648
```

Observation:

The outputs of all three pillars were successfully combined into a single integrated dataset. It contains predicted Surge Tier values, predicted CEPI scores, and geographic cluster assignments. The integrated dataset contains 40,000 records and 33 columns. Geographic cluster values are missing for 648 records because their coordinates did not pass the selected geographic filter.

Validation - Successful validation output :

```
INTEGRATED DATASET VALIDATION
----------------------------------------------

1. Dataset Shape:
(40000, 33)

2. Missing Values:
predicted_surge_tier      0
predicted_cepi_score      0
geographic_cluster      648
dtype: int64

3. Surge Tier Validation:
predicted_surge_tier
0    27875
1    10798
2     1327
Name: count, dtype: int64
Valid Surge Tier values: True

4. CEPI Score Validation:
count    40000.00
mean        38.31
std         10.32
min          8.25
25%         30.51
50%         37.11
75%         45.21
max         97.49
```

```
4. CEPI Score Validation:
count    40000.00
mean        38.31
std         10.32
min          8.25
25%         30.51
50%         37.11
75%         45.21
max         97.49
Name: predicted_cepi_score, dtype: float64
All CEPI scores between 0 and 100: True

5. Geographic Cluster Validation:
geographic_cluster
0.0    18194
1.0      436
2.0    18369
3.0      856
4.0     1497
NaN      648
Name: count, dtype: int64
All assigned clusters valid: True

6. Duplicate Records:
Duplicate rows: 0

VALIDATION COMPLETED
```

Observation:

The integrated dataset was checked to verify the correctness of the generated predictions. All Surge Tier values were within the valid classes of 0, 1, and 2. All CEPI scores were within the range of 0 to 100, and all assigned geographic cluster values were between 0 and 4. No duplicate rows were found. The 648 missing geographic cluster values were expected due to geographic filtering. All validation checks were successful.

Final output - Saved CSV verification :

```
Final integrated dataset saved successfully
File location: /kaggle/working/final_integrated_predictions.csv
Total records: 40000
Total columns: 33

Saved file verification:
Shape: (40000, 33)
File size: 10.98 MB
```

Observation:

The final integrated dataset was successfully saved as final_integrated_predictions.csv. The saved file was loaded again to verify its contents. It contains 40,000 records and 33 columns, with a file size of approximately 10.98 MB. This confirms that the integrated output was successfully generated and saved.

10. Applications

The analysis performed in this project can support:

1. Understanding common taxi travel patterns.
2. Studying trip demand across different geographic areas.
3. Analyzing trip distance and duration.
4. Exploring unusual or less frequent geographic route patterns.
5. Supporting further research into transportation planning and optimization.

11. Limitations

The following limitations should be considered while interpreting the results:

1. The available dataset contained raw taxi trip CSV files but did not provide official competition training labels or a zone lookup table.
2. The Surge Tier and CEPI Score targets were constructed experimentally using selected dataset features. They are not verified official competition targets.
3. The very high classification and regression scores reflect how closely the models learned these constructed target formulas. They do not demonstrate equivalent real-world predictive performance.
4. Geographic clusters were created from pickup and drop-off coordinates because official taxi zone identifiers were unavailable.
5. The identified geographic patterns are candidate route patterns and cannot be confirmed as actual ghost corridors.

12. Conclusion

This project successfully demonstrated the use of data preprocessing, data analysis, classification, regression, and clustering techniques on NYC Yellow Taxi trip data.

A Random Forest Classifier was used for Surge Tier prediction, a Random Forest Regressor was used for CEPI Score estimation, and MiniBatchKMeans was used to identify five geographic clusters.

Finally, the outputs of all three pillars were integrated into one dataset containing 40,000 records and 33 columns. The integrated output was validated and saved successfully in CSV format.

Overall, this project provided practical experience in working with large datasets, applying machine learning algorithms, evaluating model results, and combining multiple machine learning outputs.

13. References

1. NYC Yellow Taxi Trip Data – Public Dataset.
2. Kaggle – NYC Transit Optimization & Anomaly Challenge.